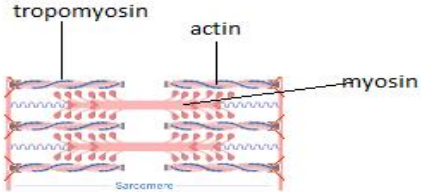


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		Biceps contract + triceps relax.	1			1		
		(ii)		Effort – bicep, load – weight, fulcrum – elbow (1) Effort is in the middle/between the fulcrum and the load/effort is between the elbow and the weight (1)		2		2		
	(b)	(i)		Any two (×1) from: Muscle respire anaerobically (1) Build-up of lactic acid/lactate (1) Lack of ATP (1)		2		2		
		(ii)		-0.05 = 2 marks 2.1 – 6.83 / 90 = 1 mark Unit = mV/s or mVs ⁻¹ (1)		3		3	2	
		(iii)		Any three (×1) from: Females fatigue {slower/ at a lower rate} (1) Overlap in data / example of overlapping data from table (1) Small sample size/ only 3 males and females (1) Unknown controlled variables e.g. Different weights / fitness/ age (1)			3	3		3
	(c)	(i)		3 correct for 1 mark 	1			1		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Any four (×1) from: - Less calcium stored in sarcoplasmic reticulum/ T tubules (1) - less calcium binds to troponin (1) - Fewer myosin binding sites exposed (1) - Less cross bridges with actin/ less myosin binds to actin (1) - Less overlap of myosin and actin (1) - Shortening the sarcomere (1) Max three if answer not comparative	3		1	4		
	(d)	(i)		Thoracic (1) Facets for articulation with ribs (1)	1	1		2		
		(ii)		Forwards/ ventrally			1	1		
		(iii)		Surgery/brace/physiotherapy		1		1		
				Question 8 total	6	9	5	20	2	3